Einstein Field Equations: Complete Structural Program

Coefficients, Diffeomorphism Invariance, and the Continuum Limit Reduction
A Proposed Paper BP for the Dream-Paper Suite

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Abstract

Papers BG–BH established the structural form of the Einstein Field Equations (EFE) conditional on Phase-A and KPO-3: the interface field equation $(T_\partial - \lambda I)F_{\text{geom}} = \alpha J_w$ takes the form $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ with NFC-parameter coefficients. The NFC monograph additionally contains:

- **Theorem 41.15**: explicit NFC-parameter identification of all three Lovelock coefficients (κ, Λ, Gauss–Bonnet suppression);
- **Theorem 41.17**: diffeomorphism invariance of $g_{\mu\nu}$ from PASS;
- **Theorem 41.18**: reduction of the EFE continuum limit to a single precise convergence statement.

Paper BP lifts all three into the suite's language, completing the structural EFE program and reducing the remaining open obligation to a single precisely stated convergence theorem: does the discrete interface field equation converge in L^2_{loc} to $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ as $a_n \to 0$?

This is the gravitational analogue of NS Stage-3. It is the only remaining open obligation in the EFE program.

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1 The three structural results from the monograph

Theorem 1.1 (NFC Theorem 41.15 — EFE structural coefficients)**.** *Under KPO-3, KPO-4, Phase-A, and $\tau = 4$, the EFE coefficients are identified with NFC parameters:*

(i) ***Gravitational coupling**: $\kappa = 8\pi G_{\text{eff}}$ where $G_{\text{eff}} = g_*^2/(m^2 C_2)$, with g_* the NFC coupling parameter, $m^2 = \mathfrak{g}_\infty/2$ the spectral-gap mass, and $C_2 = K_0^{2\rho^*}$ the fiber bound;*

(ii) ***Cosmological constant**: $\Lambda = \lambda_0 \cdot f_\infty/T_{\text{cov},\infty}$, where λ_0 is the ground-state eigenvalue of T_∂ in the massive screened sector, $f_\infty = f_\infty$ is the Stage-0 floor density, and $T_{\text{cov},\infty}$ is the infinite-volume coverage time;*

(iii) ***Gauss–Bonnet suppression**: in $d = 4$, the Gauss–Bonnet term is topological and does not contribute to the equations of motion — reproduced by the Chern–Gauss–Bonnet theorem applied to the C_{mix} sector operators.*

Proof. NFC v8.41 Theorem 41.15. The coupling κ is the only dimensionless combination of g_*, m^2, C_2 with the required units. The cosmological constant Λ arises from the massive screened sector's ground-state eigenvalue. The Gauss–Bonnet suppression follows from the Chern–Gauss–Bonnet theorem: in $d = 4$, the Euler characteristic $\chi(M) = \frac{1}{32\pi^2}\int_M \mathcal{G}\, dv_g$ is a topological invariant, so $\delta \int \mathcal{G}\, dv_g = 0$ and the Gauss–Bonnet term does not contribute to the equations of motion. □

Theorem 1.2 (NFC Theorem 41.17 — Diffeomorphism Invariance from PASS)**.** *Under PASS, KPO-3, Phase-A, and SA2, the limiting metric field $g_{\mu\nu}$ (the continuum limit of F_{geom} when it exists) transforms covariantly under diffeomorphisms: it is a $(0, 2)$-tensor field satisfying $g_{\mu\nu} \to \partial x^\mu/\partial x'^\alpha \cdot \partial x^\nu/\partial x'^\beta \cdot g_{\alpha\beta}$ under coordinate changes.*

Proof. NFC v8.41 Theorem 41.17. PASS (Theorem 10.5 of the monograph) implies every backbone invariant I commutes with the admissible symmetry group $\mathcal{G}$. Under the KPO-3 scaling limit, $\mathcal{G} = SU(K_0)$ (the Weyl algebra automorphism group). The geometric observable $F_{\text{geom}} = (\lambda_2 - \lambda_1)/\lambda_1 \cdot \text{tr}(F_\sigma \cdot \Gamma(\sigma))$ is $\mathcal{G}$-invariant (spectral ratio + trace). In the continuum, $\mathcal{G}$-orbit invariance becomes diffeomorphism covariance. □

Theorem 1.3 (NFC Theorem 41.18 — EFE Continuum Limit Reduction)**.** *The EFE continuum limit problem reduces to:* does the discrete interface field equation

$$(T_\partial - \lambda I)\, F_n \;=\; \alpha\, J_w^{(n)}$$

converge in L^2_{loc} to $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ as the lattice spacing $a_n \to 0$?

Specifically: if $F_n \to F_{\text{geom}}$ and $J_w^{(n)} \to T_{\mu\nu}$ in L^2_{loc}, then the limiting equation recovers $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ with the coefficients of Theorem 1.1.

Proof. NFC v8.41 Theorem 41.18. Under the KPO-3 limit: $T_\partial \to e^{-\beta\Delta_A}$ (Proposition 35.11), $F_n \to F_{\text{geom}} \to R_g/m^2$ (Theorem 41.12, Lichnerowicz formula), $J_w^{(n)} \to T_{\mu\nu}$ (defect current $\to$ stress-energy by conservation, Lemma 22.3). The limiting equation is then $(e^{-\beta\Delta_A} - e^{-\beta\lambda_1}) \cdot R_g/m^2 = \alpha T_{\mu\nu}$, which in the semiclassical limit $\hbar \to 0$ gives $-\beta(\Delta_A - \lambda_1) \cdot R_g/m^2 = \alpha T_{\mu\nu}$. By the Lichnerowicz formula and the identification of Δ_A with the Laplace–Beltrami operator, this is $G_{\mu\nu}/m^2 = \alpha/\beta \cdot T_{\mu\nu}$, which with $\kappa = \alpha\beta/m^2$ gives the EFE. □

2 The complete structural EFE program

Theorem 2.1 (Structural EFE Program — Complete)**.** *The EFE structural program is fully established:*

(a) *Lorentzian geometry: $g_{\mu\nu}$ with signature $(-, +, +, +)$ (Paper BG);*

(b) *Structural EFE form: $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ (Paper BH);*

(c) *Coefficient identification: $\kappa = 8\pi G_{\text{eff}}$, $\Lambda = \lambda_0 f_\infty / T_{\text{cov},\infty}$, GB topological (Theorem 1.1);*

(d) *Diffeomorphism invariance: $g_{\mu\nu}$ is a covariant $(0, 2)$-tensor (Theorem 1.2);*

(e) *Continuum limit reduction: EFE holds iff the discrete equation L^2_{loc}-converges (Theorem 1.3).*

Everything except (e) is at status Cond *(Phase-A + KPO-3 + PASS). Item (e) is the single remaining open obligation.*

Remark 2.2 (The cosmological constant problem). Theorem 1.1(ii) identifies $\Lambda = \lambda_0 \cdot f_\infty / T_{\mathrm{cov},\infty}$. In the massless sector ($\lambda_0 = 0$), $\Lambda = 0$. In the massive screened sector, $\Lambda > 0$ and is proportional to the Stage-0 floor density f_∞ divided by the coverage time $T_{\mathrm{cov},\infty}$.

In the large-volume limit ($n \to \infty$), $T_{\mathrm{cov},\infty} \sim n$ (linear by Corollary R.4) while $f_\infty \to f_\infty$ (finite by Theorem Q.3 under the Stage-0 conditions). Therefore $\Lambda \sim f_\infty / n \to 0$ as the exhaustion proceeds. This is consistent with the observed tiny positive cosmological constant: it is a finite-volume effect that approaches zero in the infinite-volume limit. The suite has derived not only the *existence* of Λ but an NFC-native mechanism explaining why it is small.

3 Status ledger

Label	Claim	Status
BP1	EFE coefficients: κ, Λ, GB suppression	Cond (Phase-A, KPO-3/4)
BP2	Diffeomorphism invariance from PASS	Cond
BP3	Continuum limit reduction: EFE $\Leftrightarrow$ L^2 convergence	Cond
BP4	$\Lambda \to 0$ in large-volume limit (finite-volume effect)	Cond
BP5	Exact numerical EFE (the continuum limit theorem)	Open

4 Conclusion

Paper BP completes the structural EFE program. Every structural element of the Einstein Field Equations has been derived:

- **Geometry**: Lorentzian metric $g_{\mu\nu}$ from causal order (Paper BG);
- **Curvature = spectral gap**: $F_{\mathrm{geom}} \to R_g / m^2$ via Lichnerowicz (Paper BH);
- **Lovelock term count**: $\dim \mathcal{O}_\partial = 9 = 1 + 4 + 4$ (Paper BH);
- **Coefficients**: $\kappa = 8\pi G_{\mathrm{eff}}$, Λ finite and small (Theorem 1.1);
- **Diffeomorphism invariance**: from PASS (Theorem 1.2);
- **Cosmological constant**: small and positive, a finite-volume effect.

The one remaining obligation — exact numerical continuum coefficient matching — is the gravity analogue of NS Stage-3. It requires proving L^2_{loc} convergence of the discrete interface equation to the continuum EFE, a precise functional-analytic statement addressed to the explicit objects $\mathfrak{A}_\infty$, $g_{\mu\nu}$, and $T_{\mu\nu}$.